Chapter 13: Introduction to Graphs

Worksheet 13: Plotting Points, Line Graphs, and Reading Graphs

Learning Objective: Students will plot points on a Cartesian plane (all four quadrants); draw and interpret line graphs, bar graphs, and pie charts; identify linear relationships from graphs; and use graphs to solve real-life problems.

Worked Example

Plot the points $A(3, 2)$, $B(-2, 4)$, $C(-3, -1)$, $D(2, -3)$ on a Cartesian plane.
Quadrant identification: $(+, +) = \text{I}$; $(-, +) = \text{II}$; $(-, -) = \text{III}$; $(+, -) = \text{IV}$.
So A is in I, B in II, C in III, D in IV.

Questions 1–5: Easy / Recall

Q1. Name the quadrant for each point: $(-3, 5)$, $(4, -2)$, $(-1, -7)$, $(6, 8)$.

Q2. What are the coordinates of the origin?

Q3. A point on the x-axis has y-coordinate _____. A point on the y-axis has x-coordinate _____.

Q4. Plot and label: $P(0, 4)$, $Q(-3, 0)$, $R(5, -2)$ on a coordinate grid. (Teacher Verification Required)

Q5. What type of graph is best to show: (a) temperature over a week, (b) favourite colours of students?

Questions 6–10: Basic Application

Q6. A car travels at 60 km/h. Make a table of distance vs. time for $t = 0, 1, 2, 3, 4$ hours. Plot the graph. (Teacher Verification Required)

Q7. From a graph, the cost C (in Rs.) of printing n books follows $C = 50n + 200$. Find the cost for: (a) 5 books, (b) 20 books. How many books can be printed for Rs. 1200?

Q8. The points $(1, 3)$, $(2, 6)$, $(3, 9)$ lie on a line. What is the relationship between x and y? Write the equation.

Q9. A pie chart shows: School: 35%, Sleep: 33%, Play: 15%, Homework: 12%, Other: 5%. If a student has 24 hours in a day, how many hours are spent on each activity?

Q10. Find the distance between points $A(3, 4)$ and $B(-1, 1)$ using the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Questions 11–15: Practice and Skill Building

Q11. The equation $y = 2x - 1$: make a table for $x = -2, -1, 0, 1, 2$ and plot the graph. Is it a straight line? (Teacher Verification Required)

Q12. Temperature conversion: $F = \frac{9C}{5} + 32$. Complete the table for $C = 0, 20, 40, 100°\text{C}$ and plot the graph. (Teacher Verification Required)

Q13. The midpoint of a line segment joining (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Find the midpoint of $(4, -2)$ and $(-6, 8)$.

Q14. A line graph shows monthly savings: Jan Rs. 500, Feb Rs. 700, Mar Rs. 600, Apr Rs. 900, May Rs. 800. Find the mean monthly savings. In which month did savings grow the most compared to the previous month?

Q15. Does the point $(3, 7)$ lie on the line $y = 2x+1$? Does $(-2, -3)$? Verify algebraically.

Questions 16–18: Higher Order Thinking

Q16. Two lines are given: $y = x + 2$ and $y = 3 - x$. Find their intersection point algebraically. Verify by plotting both lines. (Teacher Verification Required)

Q17. The vertices of a rectangle are $A(1, 1)$, $B(5, 1)$, $C(5, 4)$, $D(1, 4)$. Find the perimeter and area using the coordinates. Verify using the distance formula.

Q18. A linear graph passes through $(2, 5)$ and $(4, 9)$. Find the equation of the line and predict the value of y when $x = 10$.

Questions 19–20: Real-Life Applications

Q19. The population of a village was recorded every 10 years: 1980: 2000, 1990: 2500, 2000: 3200, 2010: 4000, 2020: 5000. Plot a line graph. By what percentage did population grow from 1980 to 2020? (Teacher Verification Required)

Q20. A mobile phone plan charges Rs. 200 fixed and Rs. 1 per call minute. Another plan charges Rs. 100 fixed and Rs. 2 per call minute. Write equations for each plan and plot on the same graph. At how many minutes are the plans equal in cost? (Teacher Verification Required)

Reflection Question: What is the difference between a bar graph and a line graph? When would you prefer each? Give real-life examples.